

PROJECT REPORT

Auto Ticket Classification using Flow Designer

A ServiceNow Automation Project for IT Service Management

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Department	B.Sc. Computer Science
Academic Year	3rd Year
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1. Project Overview & Requirements Analysis

Problem Statement

The school IT helpdesk receives multiple incident requests daily from students and teachers, such as Wi-Fi issues, projector failures, password problems, and slow computers. Currently, IT staff manually reviews each request and assigns a category, which is time-consuming and inefficient. This project automates ticket classification using ServiceNow Flow Designer by analyzing keywords in the incident's short description and description fields.

Project Objectives

- Automatically classify incidents at the time of creation.
- Reduce manual effort and categorization overhead for IT agents.
- Improve ticket routing efficiency and mean time to resolution (MTTR).
- Implement a robust, no-code, and easily maintainable solution in ServiceNow.

Business Requirements

The automated IT Service Management system must fulfill the following functional criteria:

1. Automatically classify IT tickets based on keywords present in the issue description.
2. Assign both Category and Subcategory without requiring manual agent intervention.
3. Support dependent choice logic between Category and Subcategory fields.
4. Send an automated email confirmation notification to the caller upon ticket creation.
5. Store all ticket records in a structured, standardized custom database table.
6. Ensure high maintainability, security, and future scalability for institution-wide expansion.

Creation of Project Update Set

To track all configuration changes, data schema definitions, and automation flows in a portable container, a dedicated Update Set was created.

- Go to All menus → Search for Update Sets.
- Under System Update Sets → Select Local Update Sets.
- Click the New button on the top right corner of the page.
- Set Name: Project Update Set, State: In progress, Application: Global.
- Save the Form and click 'Make This My Current Set' under Related Links.

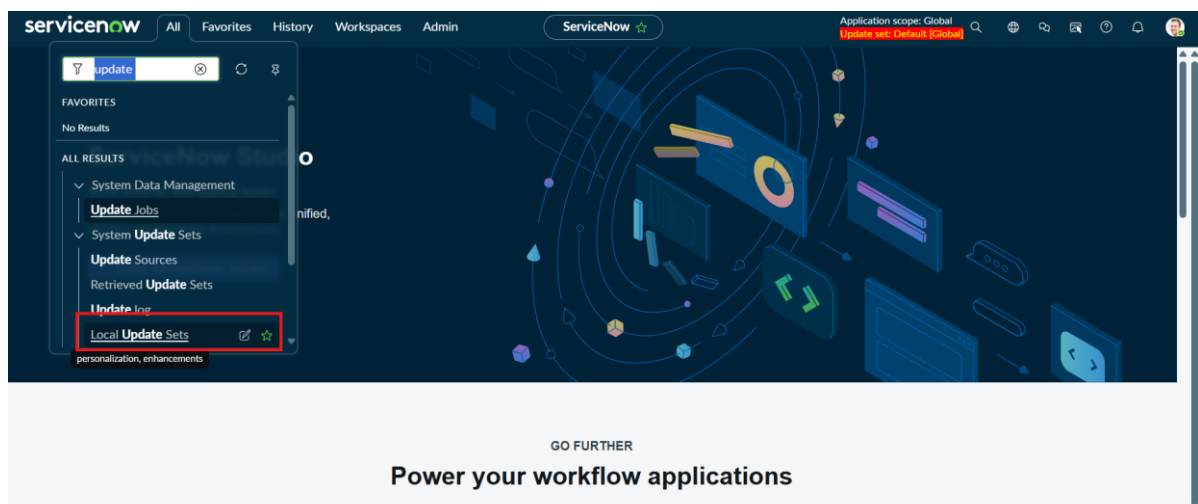


Figure 1: Local Update Set Configuration and Setup

2. Backend Development & Table Configuration

Creation of Custom Table (Incident Workflow)

A custom table was constructed under System Definition to store school incident records independently with auto-numbering enabled.

- Open All menus → Search for Tables under System Definition.
- Click the New button on the top right corner.
- Set Label: Incident WorkFlow (uncheck 'Create module' if not needed standalone).
- Save the form, navigate to Controls section, and check 'Auto-number' to True.
- Save the Form to generate system fields and auto-numbering prefix.

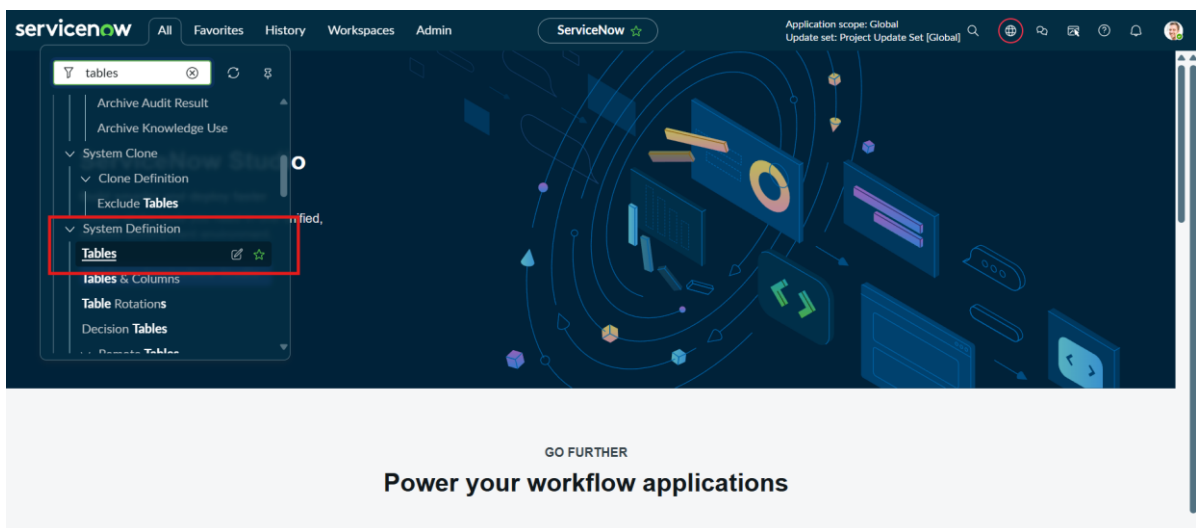


Figure 2: Custom Table Definition - Incident WorkFlow

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Logged in User	v_user_session	(empty)	false	2007-07-13 16:15:23
System Plugin	v_plugin	(empty)	false	2007-10-09 12:10:05
Active Transactions	v_transaction	(empty)	false	2008-05-23 17:30:17
Node Log	v_logfiles	(empty)	false	2009-05-14 17:05:41
Customer Uploads	v_customer_uploads	(empty)	false	2009-05-14 18:38:13
MySQL Process List	v_mysql_proclist	(empty)	false	2012-02-25 19:47:02
MySQL Status	v_mysql_status	(empty)	false	2012-02-25 19:47:08
MySQL Variables	v_mysql_variables	(empty)	false	2012-02-25 19:47:12
IOStats	v_iostats	(empty)	false	2012-02-27 17:13:51
Cluster Nodes	v_cluster_nodes	(empty)	false	2012-02-27 23:48:26
SQL Debug	v_sql_debug	(empty)	false	2012-02-28 16:40:55
Workflow Validation Report	v_wvf_validation_report	(empty)	false	2012-10-01 10:43:05
Hyper-V Resource Pool Component	cmdb_ci_hyper_v_rpool_comp	Hyper-V Object	false	2012-10-30 08:07:25
Application Menu	sys_app_application	Application File	false	2012-10-30 13:59:18
Menu Category	sys_app_category	Application File	false	2012-10-30 13:59:44
Scriptable Object	v_scriptable_object	(empty)	false	2012-11-09 15:15:23

Figure 3: Auto-Numbering Controls Configuration

Form Design & Field Creation

All required fields were created with their appropriate data types, choices, and reference targets using ServiceNow Form Designer.

Field Label	Type	Reference / Choice Values
Number	Auto Number	System Generated (e.g., INC0001)
Caller	Reference	sys_user (ServiceNow User Table)
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	Brief summary of issue
Description	String	Detailed description of issue
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

Table 1: Incident WorkFlow Table Field Specifications

Figure 4: Form Layout and Field Positioning in Form Designer

Figure 5: Choice List Configuration for Category, Subcategory, and State

3. Category-Subcategory Dependency Configuration

To ensure data consistency and prevent invalid classifications, a dynamic dependency was established between the Category and Subcategory choice fields. When a Category is selected, only the relevant Subcategory values are displayed.

Dictionary Configuration Steps

- Open the Incident Workflow table form and right-click on the Subcategory field.
- Click 'Configure Dictionary' and navigate to 'Advanced view' under Related Links.
- In the 'Dependent Field' section, check 'Use dependent field' to True.
- Set 'Dependent on field' to Category and save the record.
- Under the Choices related list, assign the corresponding Category value for each Subcategory choice.

Category	Dependent Subcategory	Targeted IT Issue Domain
Network	Wi-Fi	Wireless connectivity, router issues, IP conflicts
Hardware	Projector	Display devices, projector lamps, HDMI/VGA cabling
Access	Forgot password	Account lockouts, portal login failures, credential resets
Performance	Slow computer	System lag, memory exhaustion, application freeze

Table 2: Field Dependency Mapping Matrix

The screenshot shows the 'Dictionary Entry' configuration for the 'Subcategory' field in the 'Incident Workflow' table. The 'Dependent Field' tab is active, and the 'Use dependent field' checkbox is checked. The 'Dependent on field' is set to 'Category'. The 'Table' is 'Incident Workflow [u_incident_workflow]' and the 'Type' is 'Choice'. The 'Column label' is 'Subcategory' and the 'Column name' is 'u_subcategory'. The 'Application' is 'Global', 'Active' is checked, 'Function field' is unchecked, 'Read only' is unchecked, 'Mandatory' is unchecked, and 'Display' is unchecked. The 'Attributes' field is empty. The 'Choice List Specification' section shows the 'Dependent Field' tab selected, with 'Use dependent field' checked and 'Dependent on field' set to 'Category'.

Figure 6: Subcategory Dictionary Dependent Field Setup

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field: Category

Buttons: Create Choice List, Delete Column, Update

Related Links: [Show Table](#), [Run Point Scan](#), [\[SN Units\] Versions \(0\)](#)

Access Controls: **Choices (4)**, Attributes, Labels (1)

Search: Label Search

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-02-06 21:45:02	access
Wi-Fi	wi-fi	en	0	false	2026-02-06 21:44:50	network
Slow Computer	slow computer	en	3	false	2026-02-06 21:45:15	performance
Projector	projector	en	1	false	2026-02-06 21:45:36	hardware

Buttons: + Insert a new row...

Figure 7: Choice Values with Dependent Values Mapped

4. Automation Workflow using Flow Designer

When a user logs a support incident, they only enter their name (Caller) and the Short Description. ServiceNow Flow Designer triggers in real time upon record creation, scans the text for predefined keywords, categorizes the incident, and fires an automated email notification.

Flow Trigger Setup

- Trigger Type: Record Created
- Target Table: Incident Workflow
- Condition: Category is Empty (ensures automation executes only on unclassified tickets)

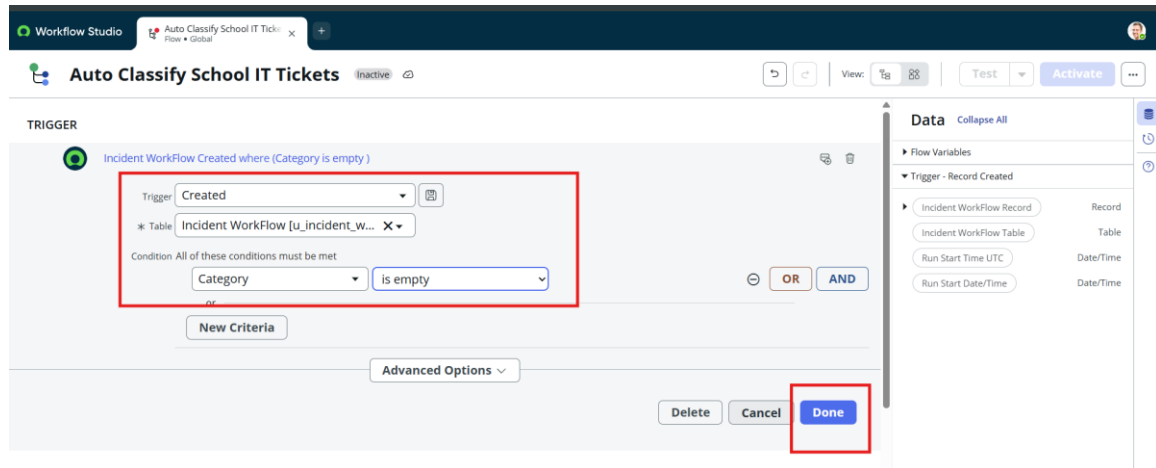


Figure 8: Flow Trigger Configuration on Record Creation

Keyword Evaluation & Classification Logic

The flow implements structured conditional logic (If / Else If) to evaluate the Short Description field:

Condition / Rule	Evaluated Keywords	Assigned Category	Assigned Subcategory
Rule 1: If (Network)	contains 'Wi-Fi' OR 'Network'	Network	Wi-Fi
Rule 2: Else If (Hardware)	contains 'Projector' OR 'Hardware'	Hardware	Projector
Rule 3: Else If (Access)	contains 'Forgot password' OR 'Password' OR 'Login'	Access	Forgot Password
Rule 4: Else If (Performance)	contains 'Slow Computer' OR 'Slow' OR 'Hanging'	Performance	Slow Computer

Table 3: Flow Designer Decision & Update Logic

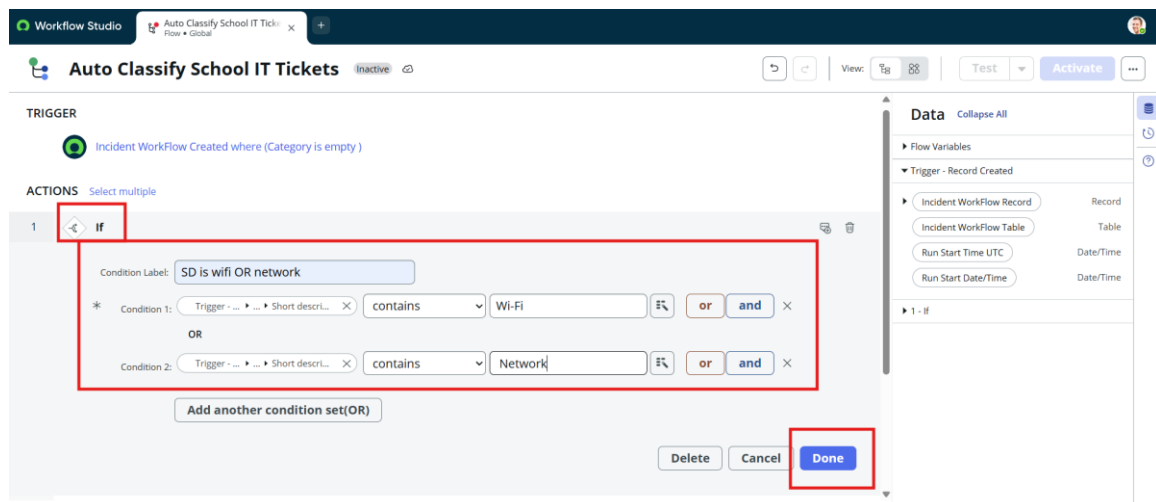


Figure 9: Wi-Fi Keyword Classification Condition & Record Update Action

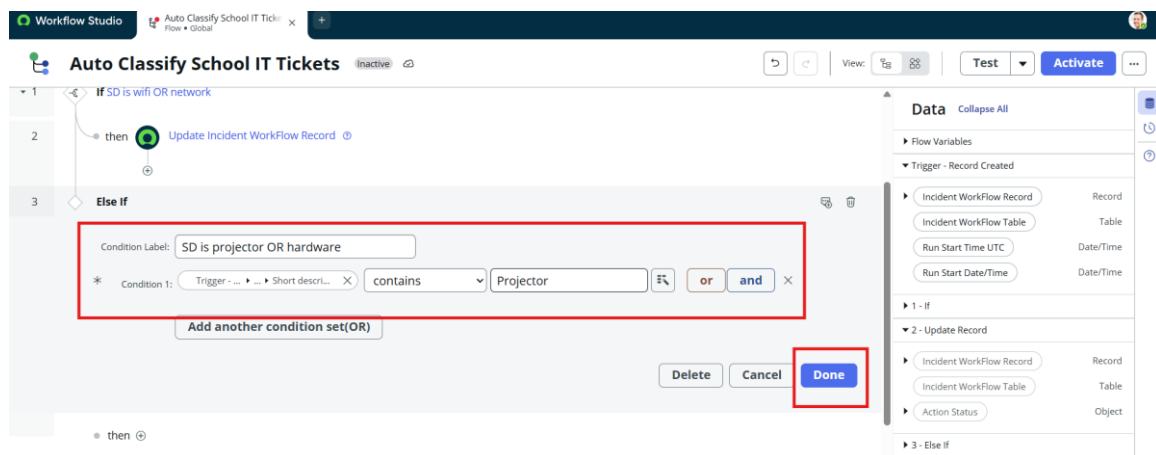


Figure 10: Projector Classification Logic & Action Inputs

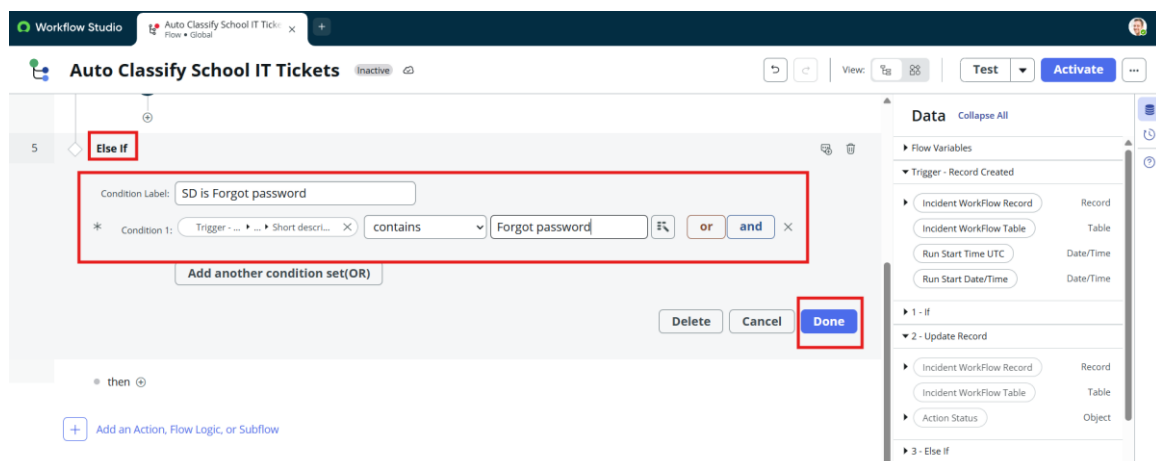


Figure 11: Password Reset Issue Classification

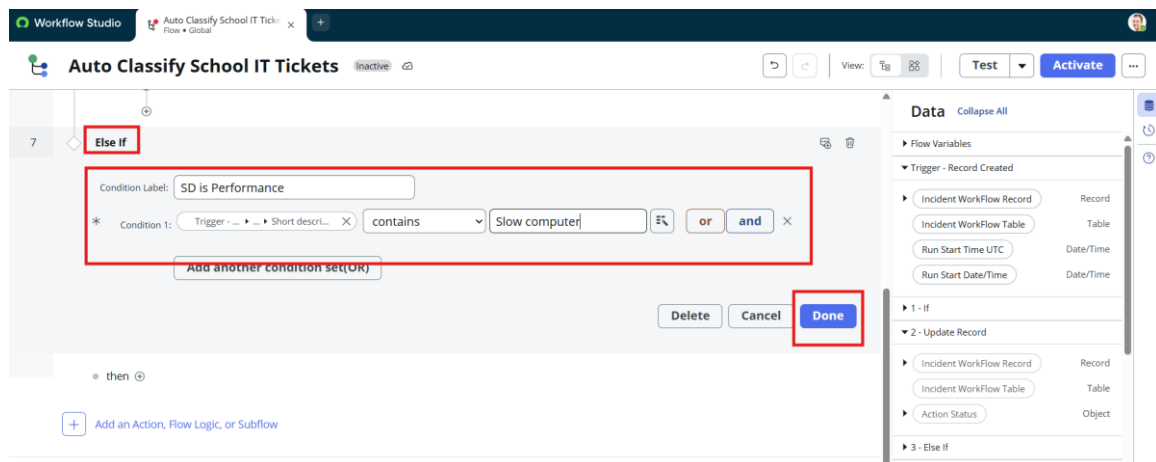


Figure 12: System Performance Issue Classification

Automated Email Notification Action

Following record classification, a 'Send Email' action is triggered to inform the caller immediately:

- Action: Send Email
- To: [Trigger -> Incident Workflow Record -> Caller -> Email]
- Subject: Your Request for the issue has been submitted
- Body: Automated confirmation notification with incident details and ticket reference number.

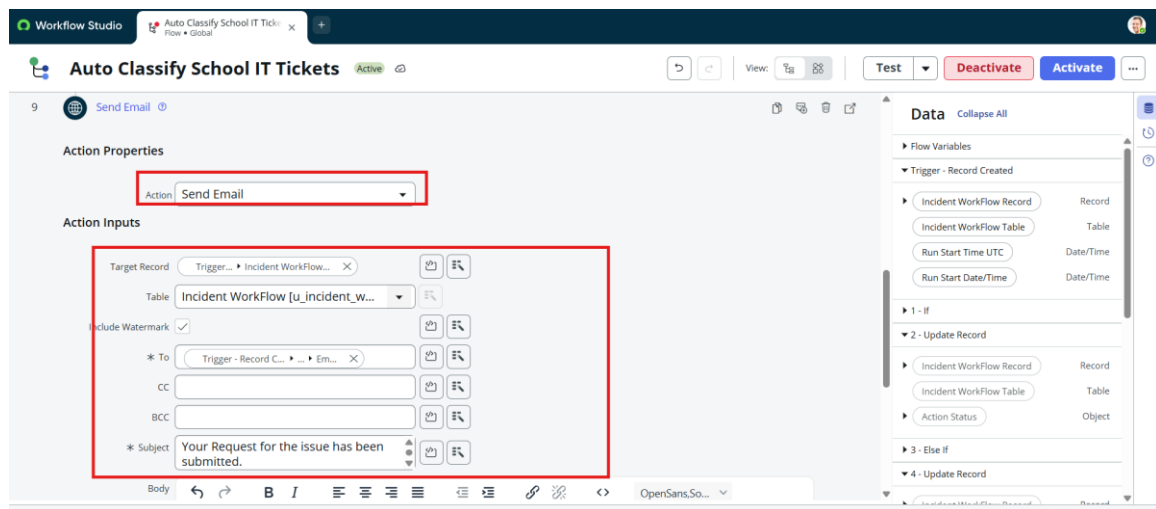


Figure 13: Send Email Action Setup in Flow Designer

5. Testing, Validation & Verification

Comprehensive end-to-end testing was conducted across multiple test scenarios to validate trigger accuracy, keyword matching precision, dependent choice integrity, and outbound notification delivery.

Test Scenarios & Results

Test Scenario	Input Short Description	Expected Category / Subcategory	Actual Result	Email Status
Scenario 1	WiFi not working in library	Network / Wi-Fi	Network / Wi-Fi	Delivered
Scenario 2	Projector not turning on	Hardware / Projector	Hardware / Projector	Delivered
Scenario 3	Forgot login password	Access / Forgot Password	Access / Forgot Password	Delivered
Scenario 4	Computer hanging and slow	Performance / Slow Computer	Performance / Slow Computer	Delivered

Table 4: End-to-End Test Case Verification Matrix

The screenshot shows the 'Incident Workflow' form in ServiceNow. The form is titled 'Incident Workflow INCO0511'. It contains the following fields:

- Number: INC00511
- Caller: Alfonso Griglen
- Category: Network
- Subcategory: Wi-Fi
- Short description: My college Has issue with Network
- Description: (empty)
- State: -- None --
- Assigned group: (empty)
- Assigned to: (empty)

There are 'Update' and 'Delete' buttons at the bottom of the form.

Figure 14: Scenario 1: Automatic Classification to Network / Wi-Fi

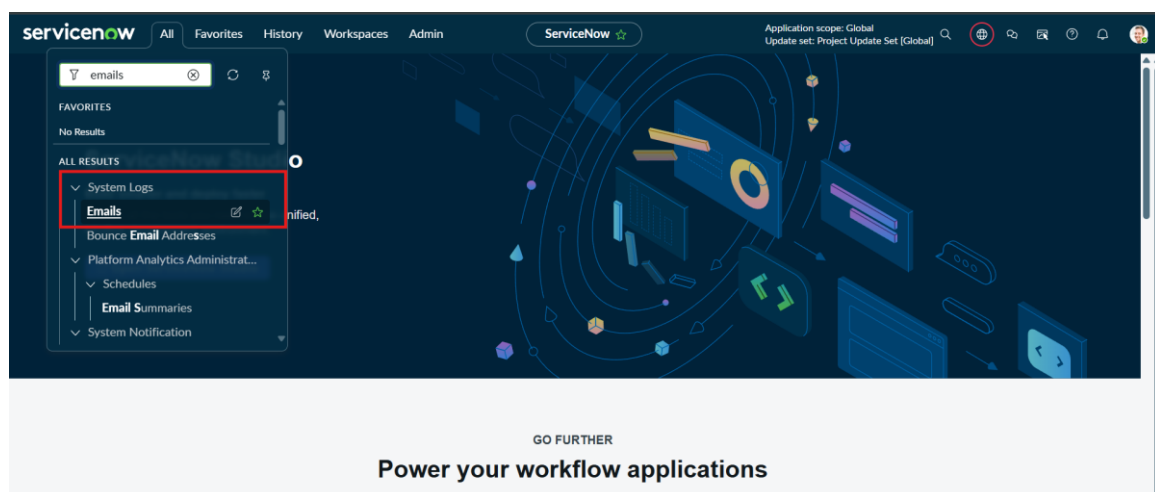


Figure 15: Email Log Verification and Outbound Message Preview

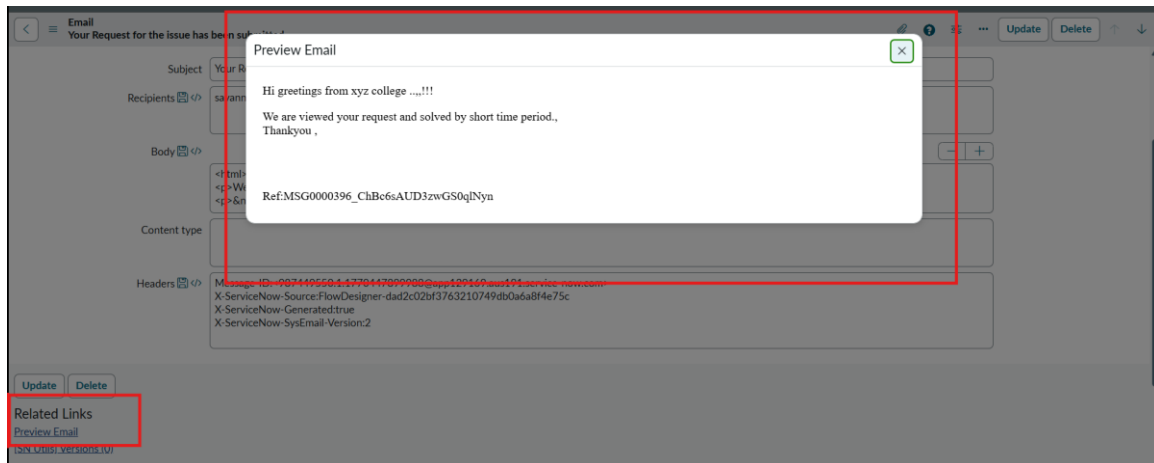


Figure 16: Scenario 2: Automatic Classification to Hardware / Projector

6. Deployment & Solution Export

To deploy the solution to target testing or production environments, the Update Set is marked Complete and exported to XML format.

- Navigate to All menus → Search for Local Update Sets under System Update Sets.
- Open 'Project Update Set' from the list.
- Change the State from 'In Progress' to 'Complete' and save the form.
- Click 'Export to XML' under Related Links to download the complete changeset file.
- The exported XML payload can be seamlessly committed to any target ServiceNow instance.

The screenshot shows the ServiceNow Incident Workflow form for incident INC00512. The form is titled 'Incident Workflow INC00512' and has a red box highlighting the main form area. The fields are as follows:

- Number:** INC00512
- * Caller:** Sam Sorokin
- Category:** Hardware
- Subcategory:** Projector
- * Short description:** Projector not turning on
- Description:** (Empty text area)

On the right side, there are additional fields:

- State:** -- None --
- Assigned group:** (Empty dropdown)
- Assigned to:** (Empty dropdown)

At the bottom left, there are 'Update' and 'Delete' buttons. At the bottom right, there is a small icon.

Figure 17: Closing and Exporting Update Set to XML

7. Conclusion & Future Enhancements

The Auto Ticket Classification using Flow Designer project successfully delivers a complete, end-to-end automation solution for the school IT helpdesk at The Madurai Diraviyam Thayumanavar Hindu College. The primary objective of eliminating manual ticket categorization and improving response time has been fully achieved.

Key Benefits Achieved

- **Zero-Touch Categorization:** Eliminates manual triage for standard IT incident requests.
- **Instant Stakeholder Communication:** Immediate email confirmation ensures transparent communication.
- **High Maintainability:** 100% no-code implementation using Flow Designer enables easy modifications without developer scripting.
- **Data Integrity:** Field dependencies ensure clean, standardized data capture across all records.

Future Scope

- **Integration with ServiceNow Predictive Intelligence and Natural Language Understanding (NLU)** for complex queries.
- **Automated assignment group and agent routing** based on technician availability and skill matrix.
- **Service Level Agreement (SLA) milestone tracking and automated escalation workflows.**